

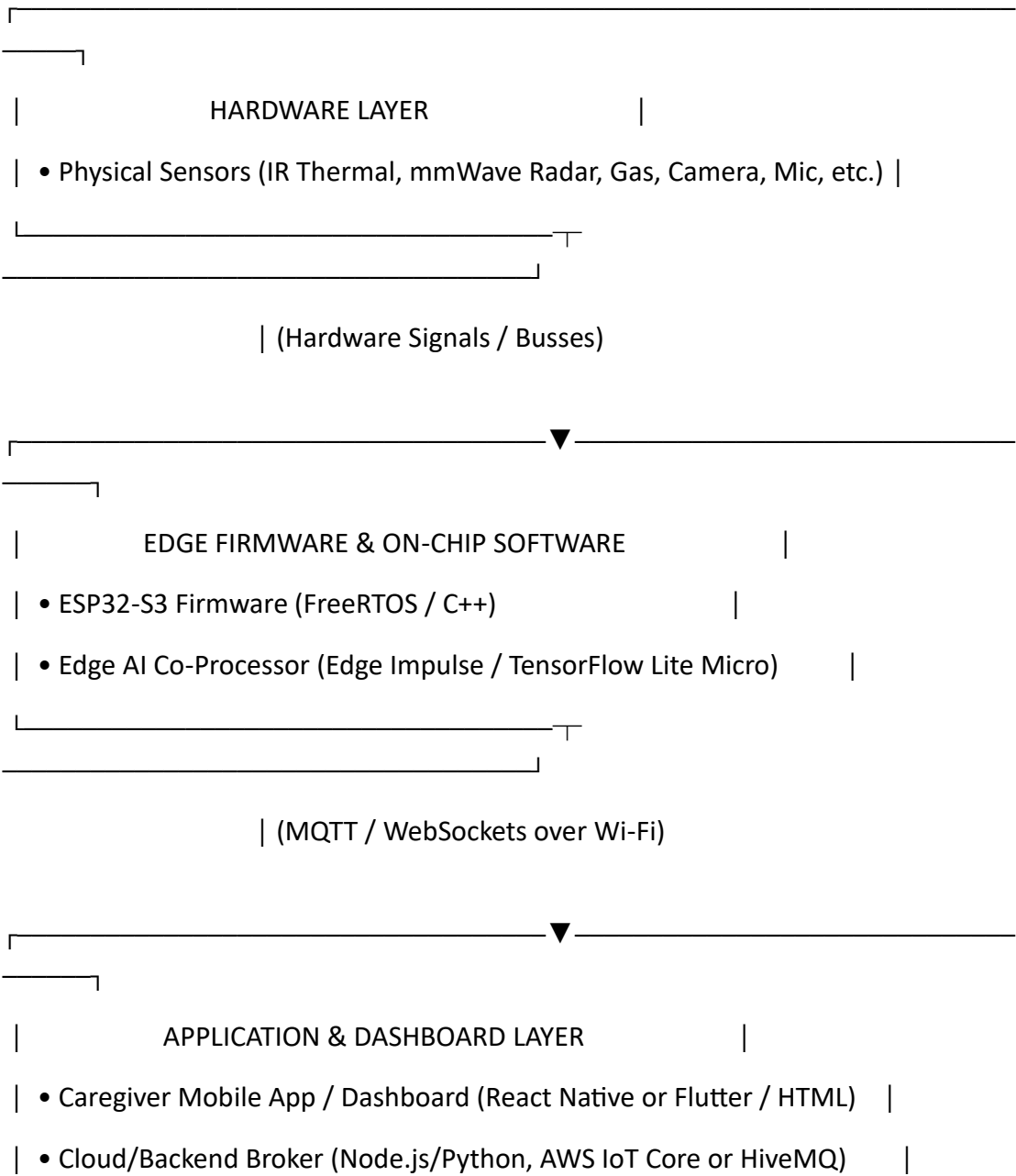
NestGuard System Architecture & Vendor Engineering Specification

Document Version: 1.0

Target Release: Enterprise / IoMT Ready

Subject: Hardware-Software Mapping, Edge AI Pipelines, and System Deployment Matrix

1. High-Level System Architecture



2. Technical Node Mapping & Deployment Specifications

Section 1: Physical Sensors & Transducers (Hardware Core)

- **IR Thermal Array (MLX90640)**
 - **Hardware:** 32×24 Far-Infrared Sensor Array.
 - **Software Stack:** Adafruit_MLX90640 C++ Driver / Linux i2c-dev library.
 - **Target Deployment:** Compiles directly into the primary ESP32-S3 firmware binary.
- **60GHz mmWave Radar (BGT60TR13C)**
 - **Hardware:** Radar SoC IC mounted on overhead housing.
 - **Software Stack:** Radar MCU SDK / C-based UART frame parser.
 - **Target Deployment:** Executes as a dedicated real-time task on the FreeRTOS scheduler.
- **Gas & Environmental Sensor (BME688 / MQ Series)**
 - **Hardware:** Metal-oxide gas sensor for VOC and NH_3 detection.
 - **Software Stack:** Bosch BSEC (Bosch Sensortec Environmental Cluster) library.
 - **Target Deployment:** Compiled as a static binary library into the ESP32 firmware.
- **Ambient Environment Sensor (BME280 / DHT22)**
 - **Hardware:** Temperature and relative humidity module.
 - **Software Stack:** Adafruit_BME280 or C-driver wrapper.
 - **Target Deployment:** ESP32 firmware task executing on a 1 Hz polling loop.
- **Mattress Load Cells (HX711) & 6-Axis IMU (MPU6050)**
 - **Hardware:** $4 \times$ corner strain gauge load cells with HX711 ADC + 6-DOF IMU under mattress.
 - **Software Stack:** HX711-multi C++ library & MPU6050_light driver.

- **Target Deployment:** ESP32 firmware running a continuous weight-fusion algorithm.
- **RGB / NIR Camera Module (OV2640 / OV5640)**
 - **Hardware:** Camera module with IR cut filter connected via DVP or MIPI-CSI.
 - **Software Stack:** esp32-camera component / Linux V4L2 drivers.
 - **Target Deployment:** Edge AI Co-Processor or Linux Micro-SBC interface.

Section 2: Communication Protocols & Edge Models

- **Thermal Processing Unit (Heat-Grid Calibration Model)**
 - **Software Stack:** C++ Matrix math library / NumPy for linear interpolation.
 - **Target Deployment:** Executes on the Edge Hub to upscale raw 32×24 frames into a 128×96 thermal overlay matrix.
- **On-Chip Phase Classifier (1D-CNN Time-Series Model)**
 - **Software Stack:** TensorFlow Lite for Microcontrollers (TFLM) / Edge Impulse C++ SDK.
 - **Target Deployment:** Flashed into internal flash memory of the ESP32-S3 or Radar DSP.
- **Audio Analysis Loop (Mel-Spectrogram CNN)**
 - **Software Stack:** CMSIS-DSP (real-time FFT) + Quantized TFLM Audio Classifier.
 - **Target Deployment:** Runs directly on the ESP32-S3 DMA Audio Pipeline via I2S.
- **Bus Protocols & Low-Level Drivers**
 - **Software Stack:** FreeRTOS drivers (i2c_master, i2s_driver, uart_driver).
 - **Target Deployment:** Low-level board support package (BSP) compiled into .bin firmware.
- **Stillness & Weight Fusion Gate (Load + IMU)**
 - **Software Stack:** Custom C++ Kalman Filter / Moving Average algorithm.

- **Target Deployment:** ESP32 firmware task maintaining a 30-second verification window.

Section 3: Central Hub & Edge AI Vision Core

- **Vision Core (MobileNet Face & Obstruction Tracking)**
 - **Software Stack:** OpenCV + TensorFlow Lite Micro / MobileNetV2 SSD runtime.
 - **Target Deployment:** Runs on dedicated Edge AI Hardware (ESP32-S3 vector core or Linux Micro-SBC like K210/Jetson).
- **ESP32 Central Processing Hub (Event Aggregator & Local Alert Logic)**
 - **Software Stack:** C++ / FreeRTOS, esp-mqtt, esp-websocket, cJSON.
 - **Target Deployment:** Main MCU flash memory; acts as the primary local event aggregator and alert orchestrator.

Section 4: Caregiver Interface & Cloud Gateways

- **Backend Communication Gateway**
 - **Software Stack:** MQTT Broker (EMQX or AWS IoT Core) + Node.js / Python FastAPI server.
 - **Target Deployment:** Cloud Container (Docker / AWS ECS / DigitalOcean) or local edge gateway.
- **Caregiver Application / Executive Dashboard**
 - **Software Stack:** React Native / Flutter (Mobile) or HTML5 / Tailwind CSS / WebSockets (Web).
 - **Target Deployment:** iOS App Store / Google Play Store / Hosted Cloud CDN (Vercel / Netlify).

3. Summary Deployment Matrix for Vendor Delivery

Functional Node	Target Hardware / IC	Software Framework	Delivery Artifact
Sensors & Transducers	MLX90640, BME688, HX711, INMP441, MPU6050	C++ Hardware Drivers / BSP	Firmware .bin file flashed on MCU

Functional Node	Target Hardware / IC	Software Framework	Delivery Artifact
Edge Micro-Models	ESP32-S3 / Edge Processing Hub	TensorFlow Lite Micro / CMSIS-DSP	Quantized .tflite model embedded in C++ headers
Vision Core	Camera Module + AI Coprocessor	OpenCV + MobileNetV2	Executable binary / C++ container on Edge SoC
Central Processing Hub	ESP32-S3 Board	FreeRTOS / ESP-IDF C++	Firmware main binary (main.bin)
Caregiver Interface	Smartphone / PC Web Browser	React Native / HTML5 + WebSockets	.apk / .ipa mobile packages or Hosted Web App